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MODEL SELECTION AND VALIDATION REPORT ( only\_kde\_feve\_biomass model )

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## OBSERVATIONS:3382

## SANITY: OK - all models in the selection process passed sanity checks.

## SELECTED RANDOM TERM: spatiotemporal\_random\_fields\_ar1

## MODEL SUMMARY ( only\_kde\_feve\_biomass model )

Spatiotemporal model fit by REML ['sdmTMB']

Formula: fixed\_formula

Time column: year

Family: lognormal(link = 'log')

Conditional model:

|               | coef.est | coef.se |
|---------------|----------|---------|
| (Intercept)   | 7.11     | 0.19    |
| kde_feve      | -0.33    | 0.02    |
| sbt_mean      | 0.26     | 0.06    |
| sic_mean      | 0.04     | 0.05    |
| sbt_sd        | 0.09     | 0.03    |
| sst_sd        | 0.05     | 0.03    |
| log_chla_mean | 0.02     | 0.02    |
| depth         | -0.02    | 0.03    |

Dispersion parameter: 0.78

Spatiotemporal AR1 correlation (rho): 0.90

Matérn range: 307.66

Spatiotemporal marginal AR1 SD: 0.80

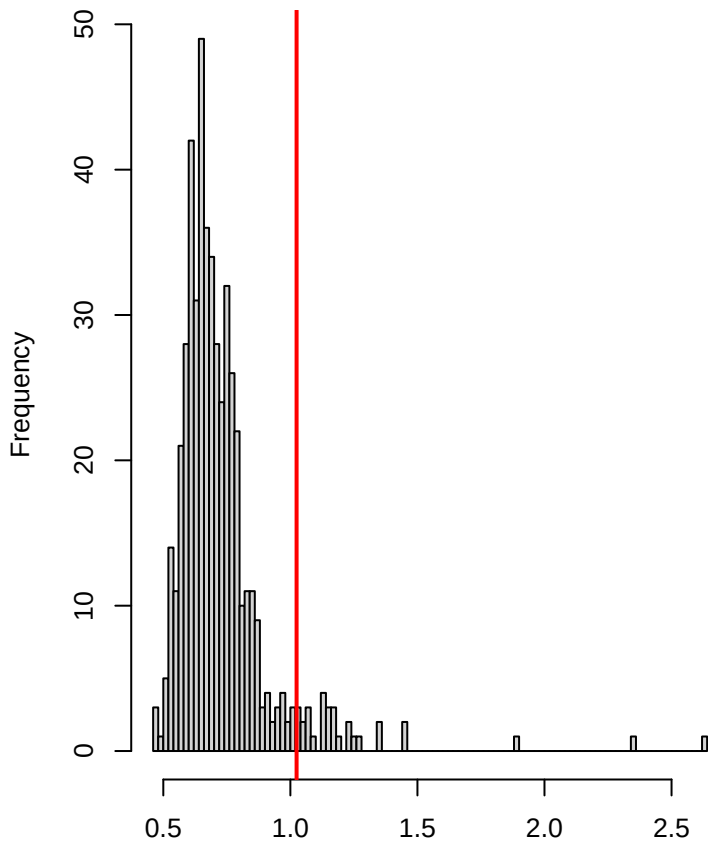
REML criterion at convergence: 27159.871

See ?tidy.sdmTMB to extract these values as a data frame.

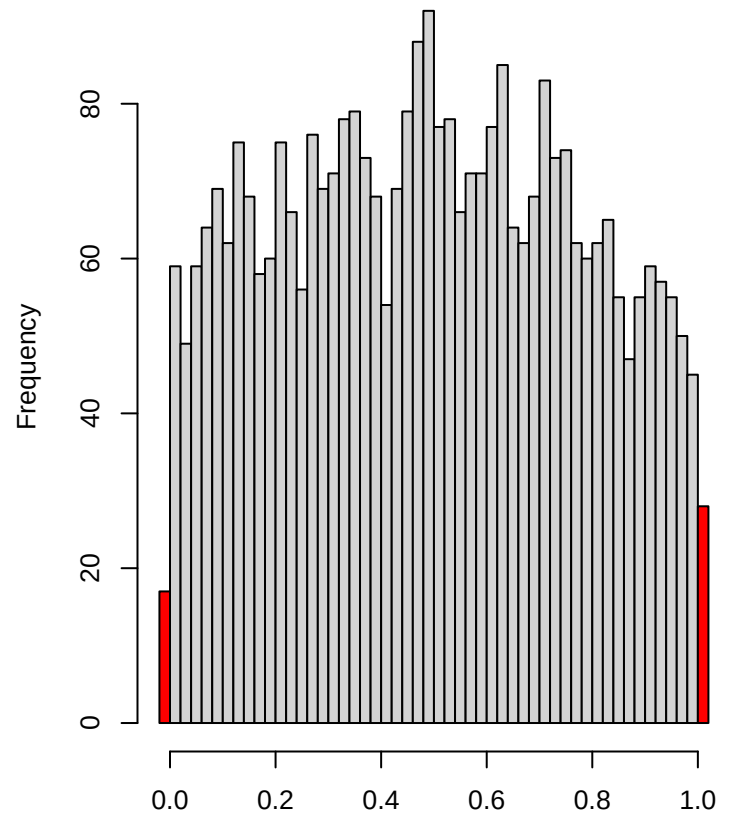
## SELECTION STATISTICS

|                                                  | df | AIC      | delta_AIC  |
|--------------------------------------------------|----|----------|------------|
| spatiotemporal_random_fields_ar1                 | 12 | 54343.74 | 3.200444   |
| spatiotemporal_random_fields_ar1_with_anisotropy | 14 | 54346.94 | 254.293088 |
| spatiotemporal_random_fields_iid                 | 11 | 54601.24 | 3.416287   |
| spatiotemporal_random_fields_iid_with_anisotropy | 13 | 54604.65 | 185.629812 |
| spatial_random_fields                            | 11 | 54790.28 | 3.493653   |
| spatial_random_fields_with_anisotropy            | 13 | 54793.77 | 577.078052 |
| error_only                                       | 9  | 55370.85 | NA         |

DHARMA nonparametric dispersion test via sd of residuals fitted vs. simulated



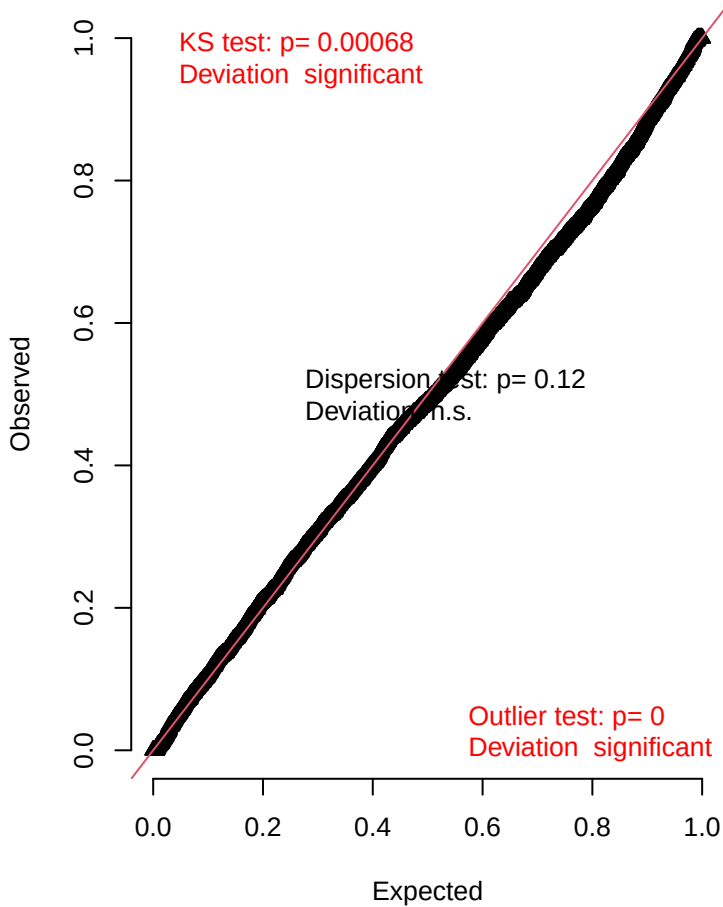
Outlier test significant



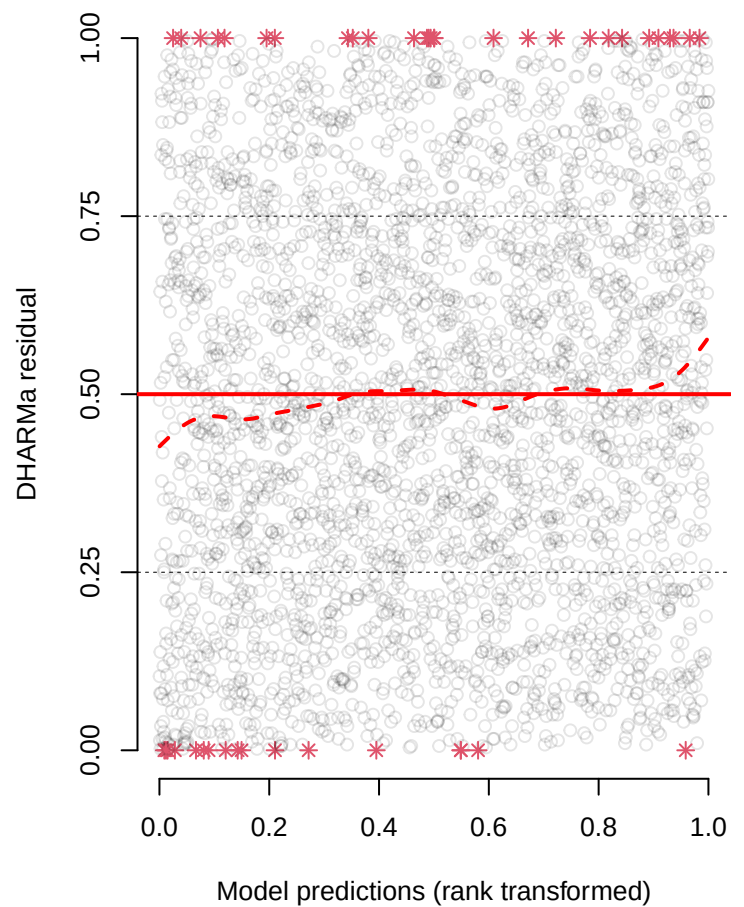
Simulated values, red line = fitted model. p-value (two.sided) :

Residuals (outliers are marked red)

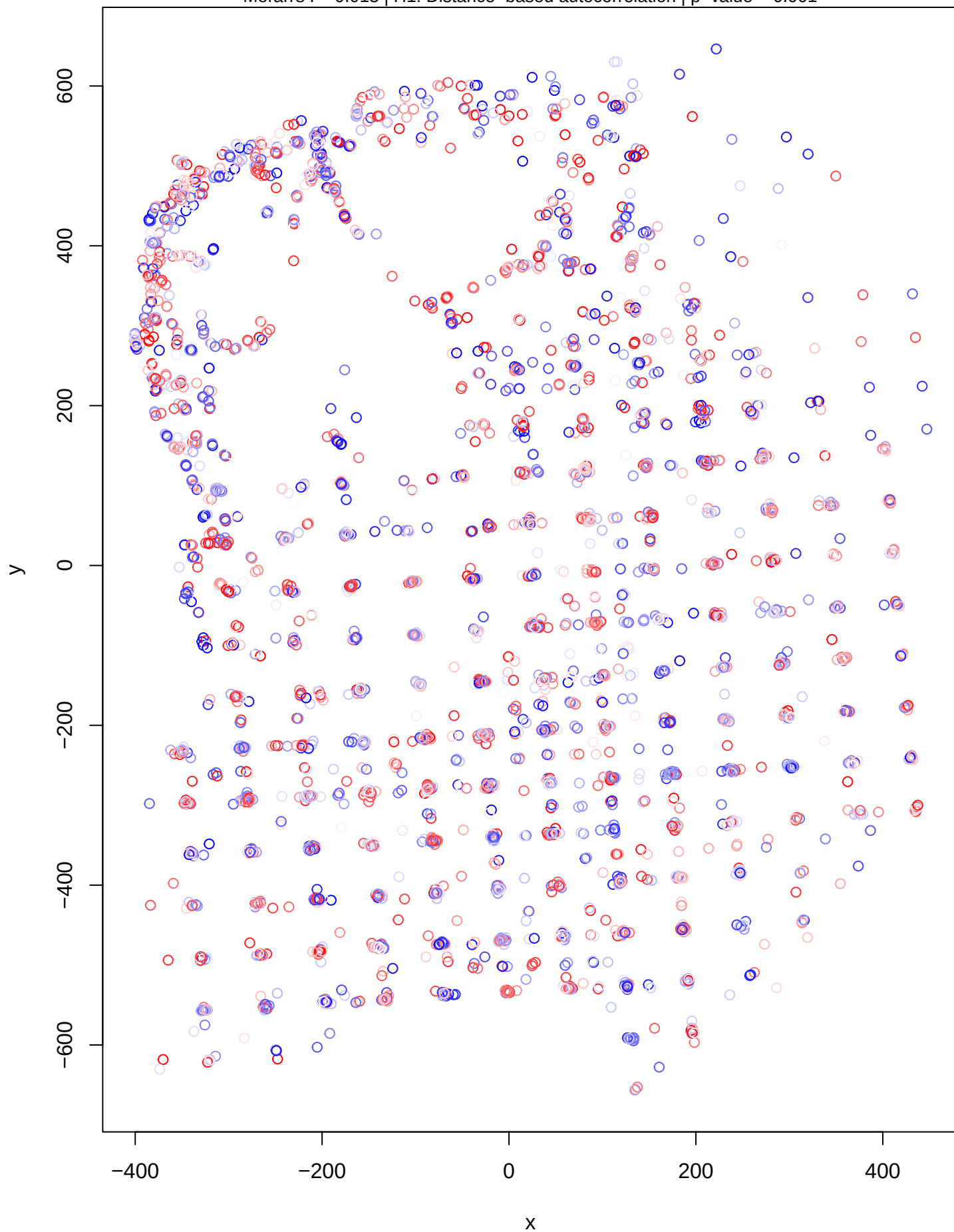
QQ plot residuals



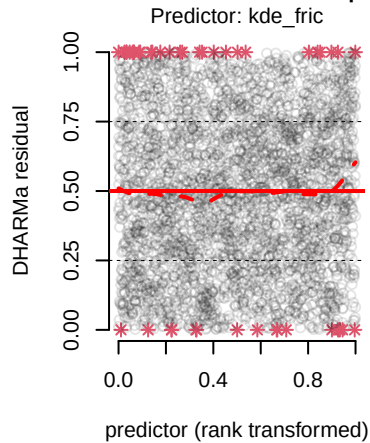
DHARMA residual vs. predicted



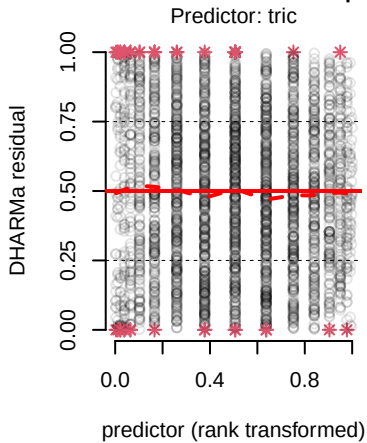
**DHARMa Moran's I test for distance-based autocorrelation**  
Moran's I = 0.013 | H1: Distance-based autocorrelation | p-value = 0.001



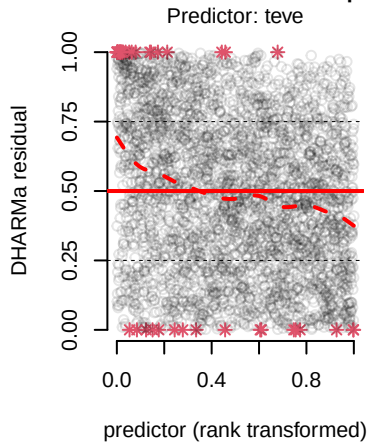
**DHARMa residual Residual vs. pred**



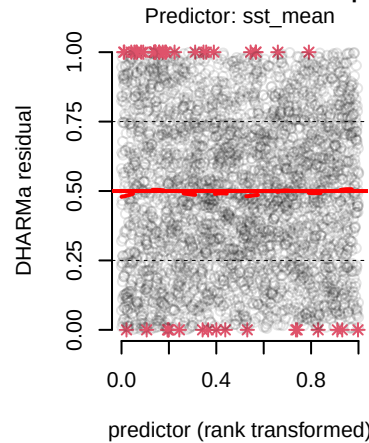
**DHARMa residual Residual vs. pred**



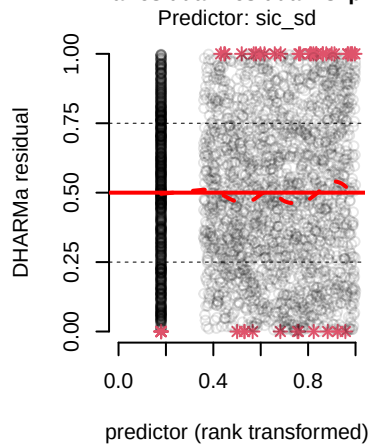
**DHARMa residual Residual vs. pred**



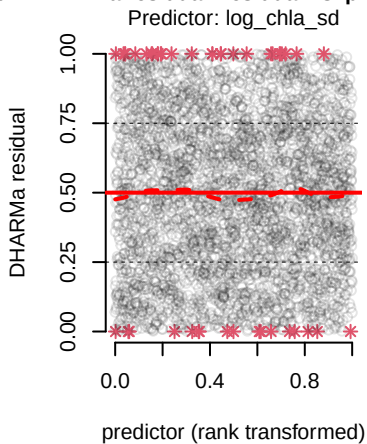
**DHARMa residual Residual vs. pred**



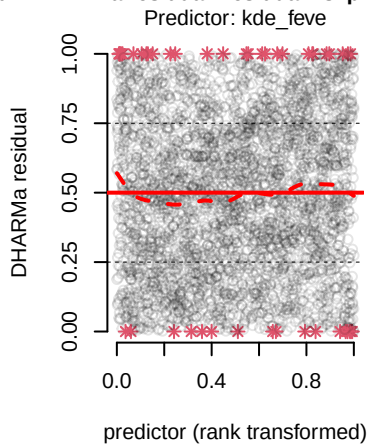
**DHARMa residual Residual vs. pred**



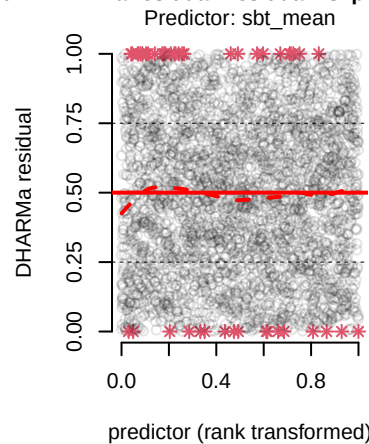
**DHARMa residual Residual vs. pred**



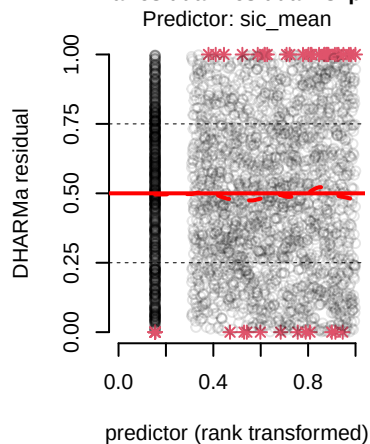
**DHARMa residual Residual vs. pred**



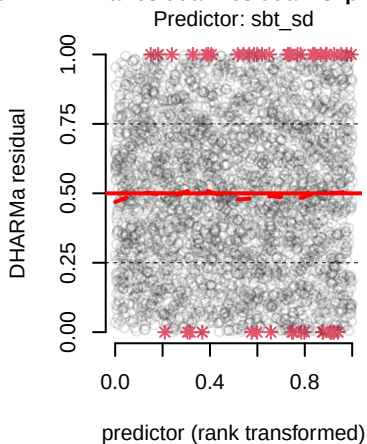
**DHARMa residual Residual vs. pred**



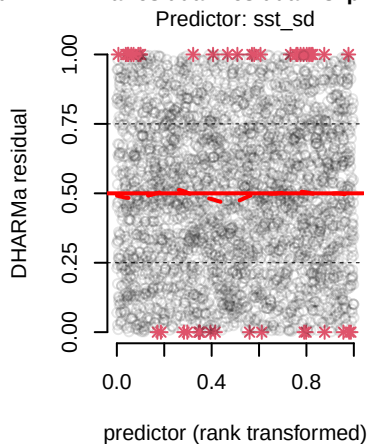
**DHARMa residual Residual vs. pred**



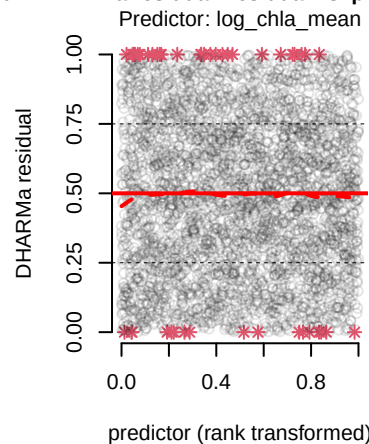
**DHARMa residual Residual vs. pred**



**DHARMa residual Residual vs. pred**



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